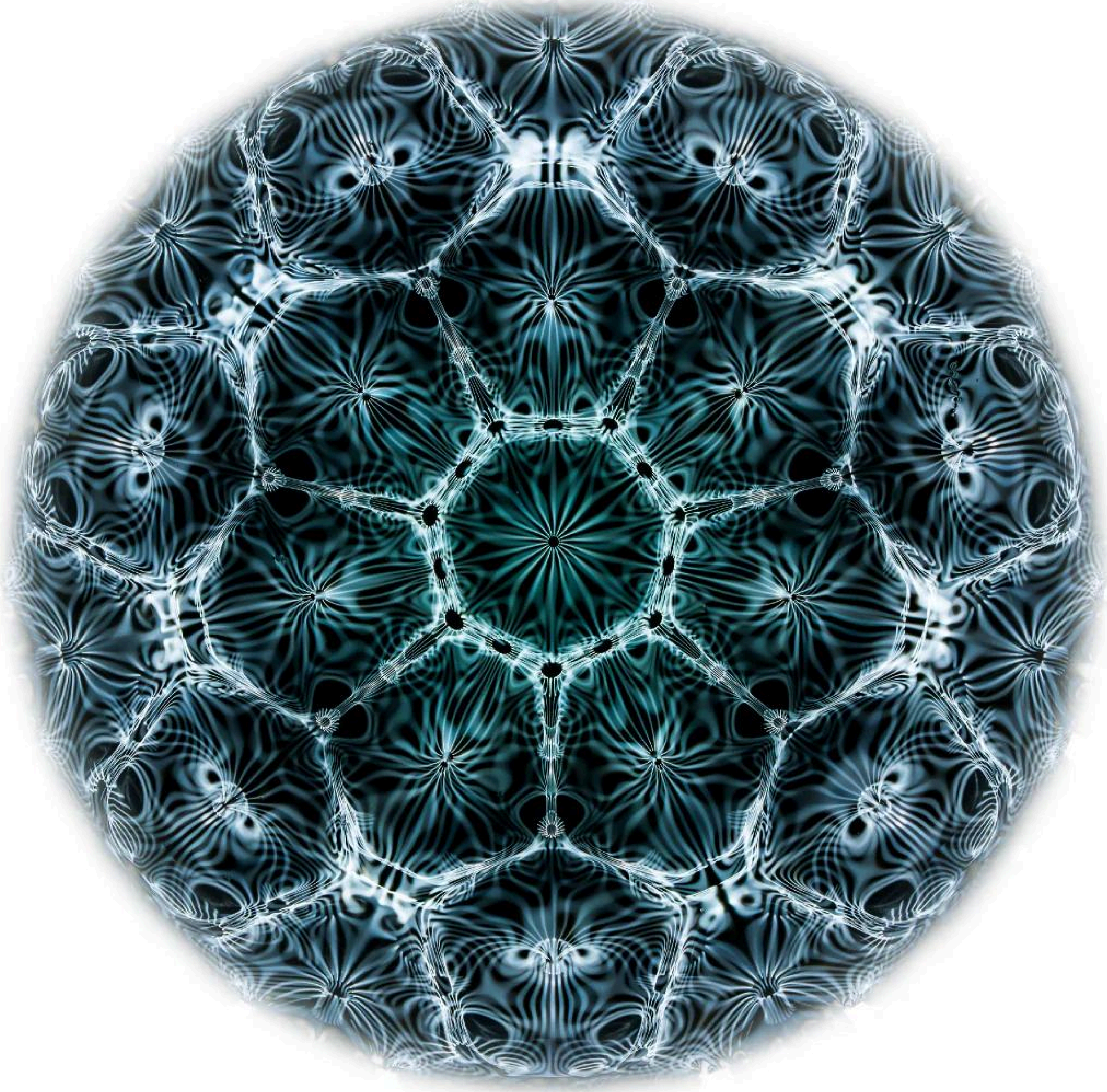
Cellular Cymatics - Using Sound to Build Biological Structures in Space

SPACE CARGO UNLIMITED

STAR HELIX

APHELION INDUSTRIES

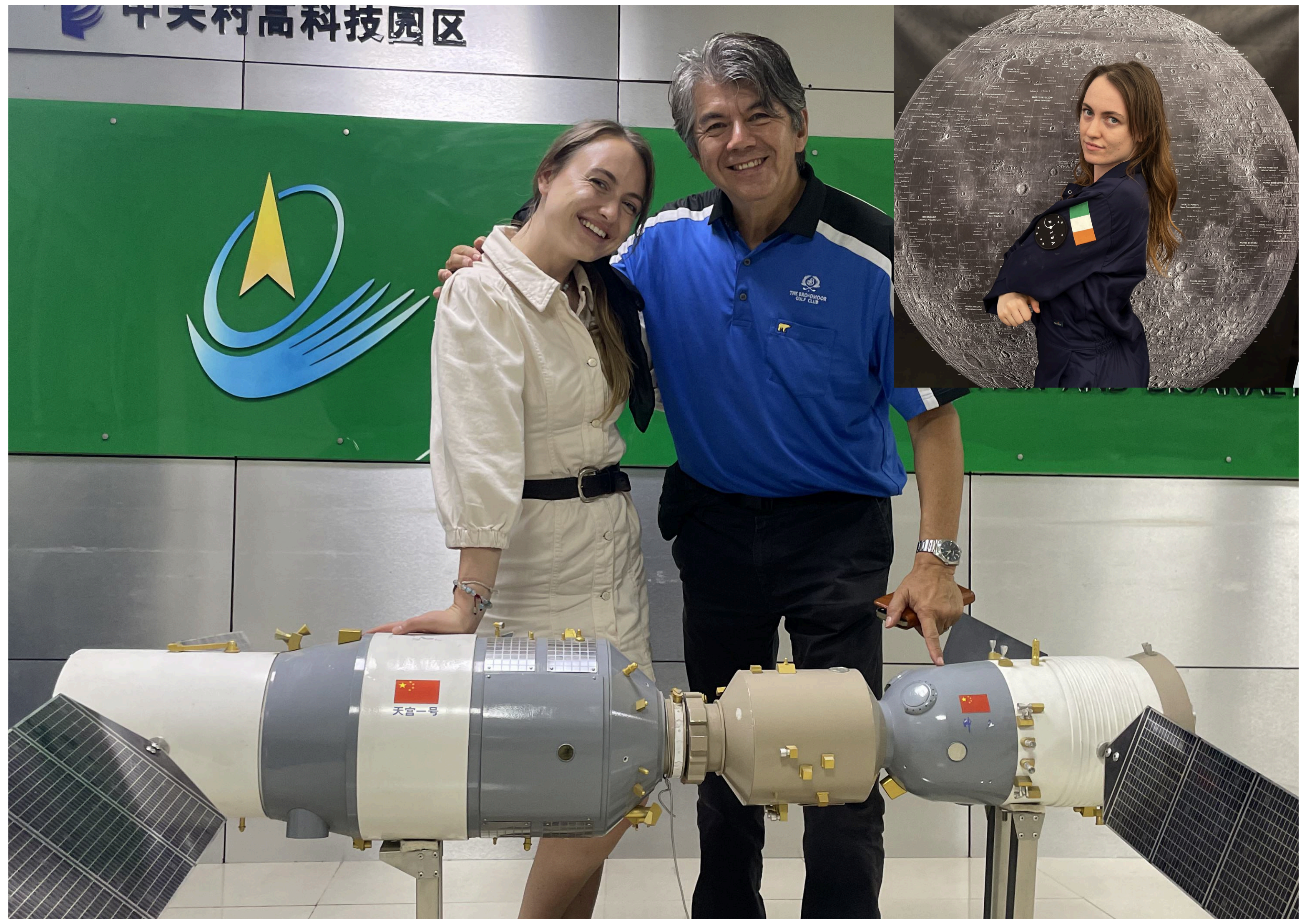
BioHackathon Edinburgh



Your mentors ||

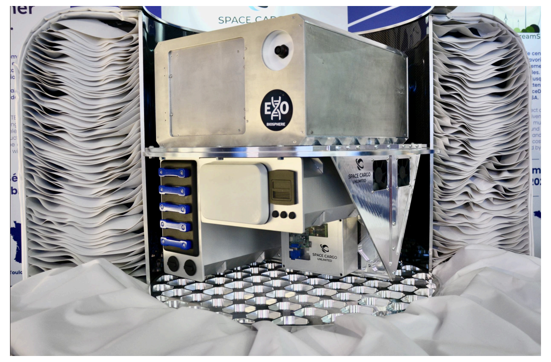
Dr.Mason Robbins of Star Helix.

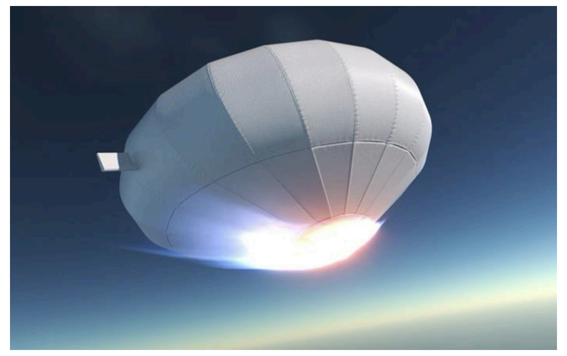
Innovating in biotech and engineering for humans to live and work safely in space.



The artist formerly known as *Binary-Pulsar Star System Wished To Be A Girl And Came Down*

SPACE CARGO UNLIMITED





Aphelion Industries CEO, ultra-resilient spacecraft.

Futurist, psychology graduate, hungry.

The challenge || Cymatics is the study of vibration, from Greek 'KYMA', meaning wave.

[=] Design Proposal:

output

A **test-rig concept** to explore the use of sound/vibration and biology to build in microgravity.

Question
Can you express three-dimensional control using vibrations?

This might consist of three elements:
+ Container Design
+ Frequency Input System (transducer)
+ Biochemical Agent-Substrate Matrix.

[=] Digitalisation / Translation:

output

A software/mathematical relationship between **frequency** and **shape**.

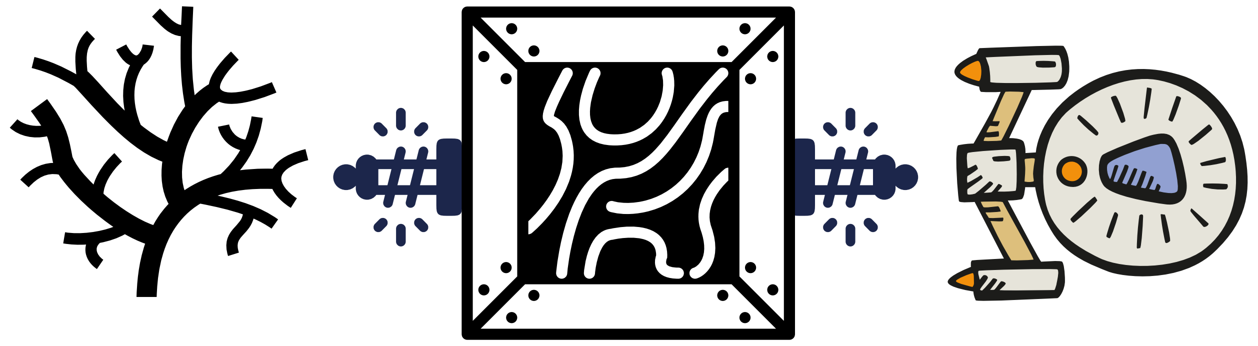
Question
Can you sample various frequencies using the rig to discover which may present useful morphologies?

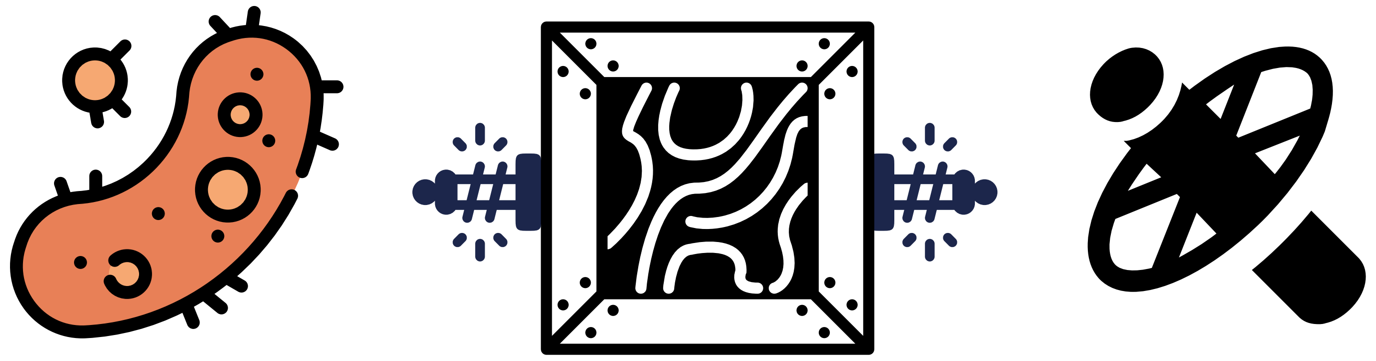
Can you link potentially useful resonant structures with particular use-cases?
Eg: in engineering, biology, architecture, etcetc

Technology Vision

In the future . . . we could **use sound**
. . . as a **blueprint** to grow things **in space!**

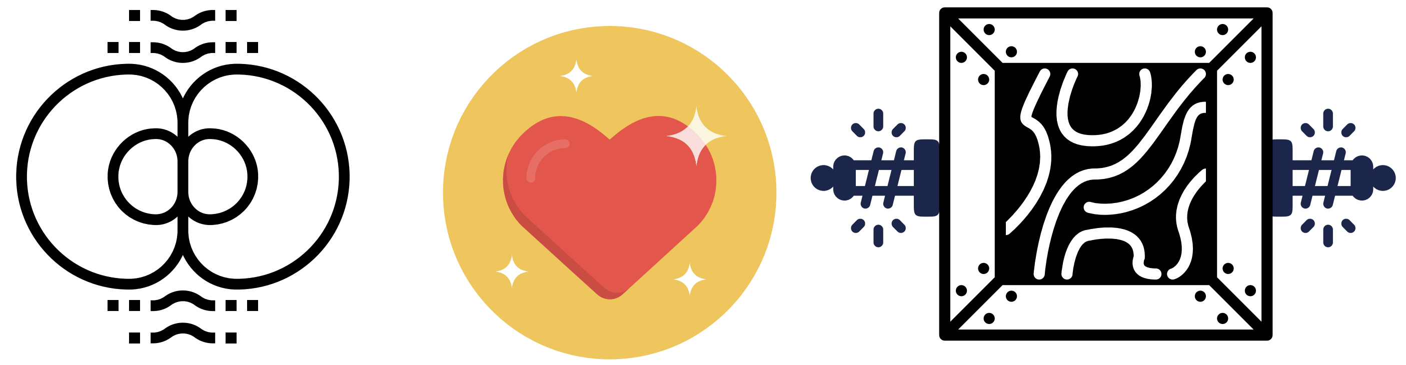
Structure a coral pathway by vibrating a box of **bio-mixture** to grow a coral spaceship?





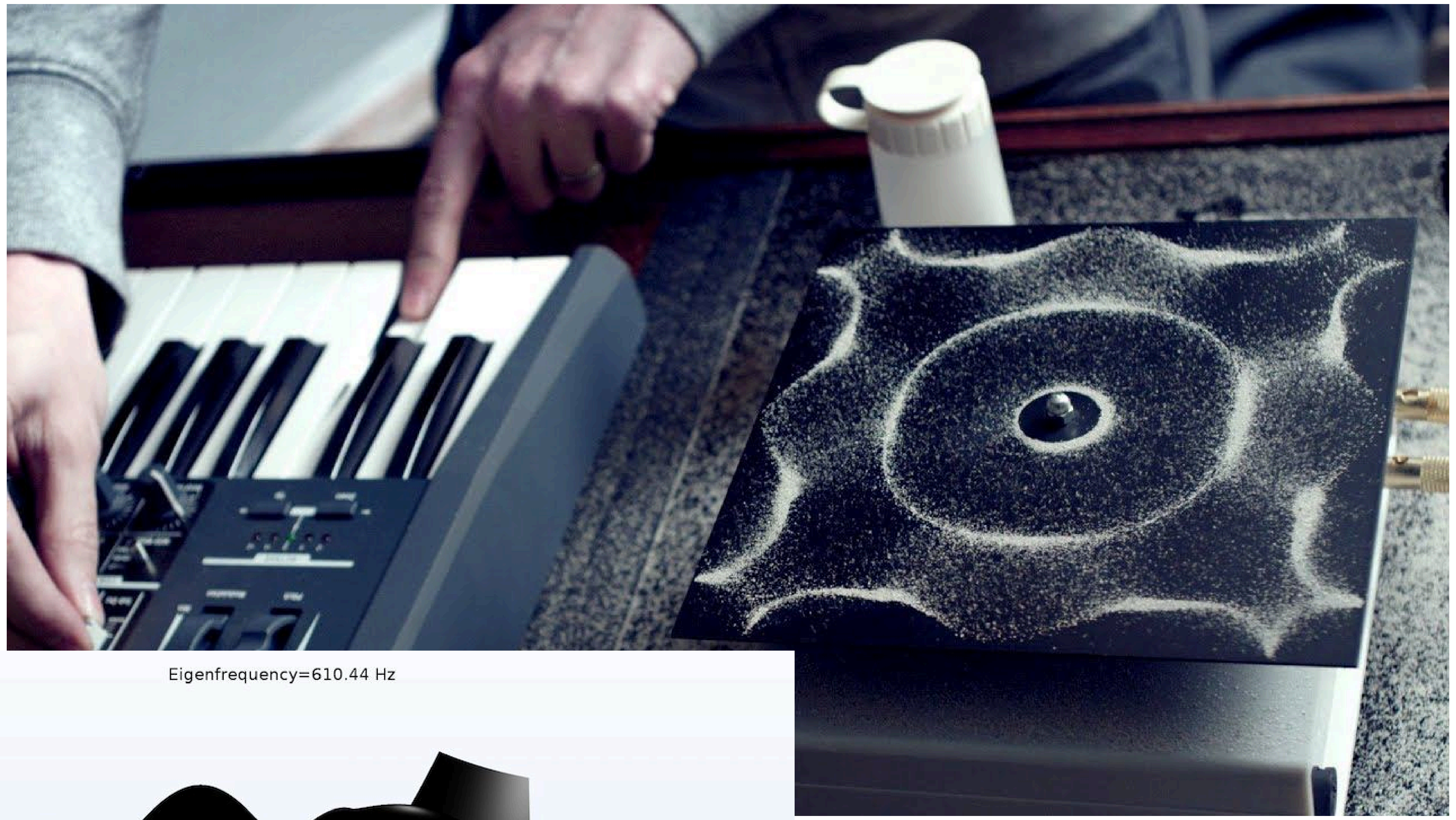
Suspend iron-mining bacteria in a **sonic-blueprint**, then sinter them to build girders in space?

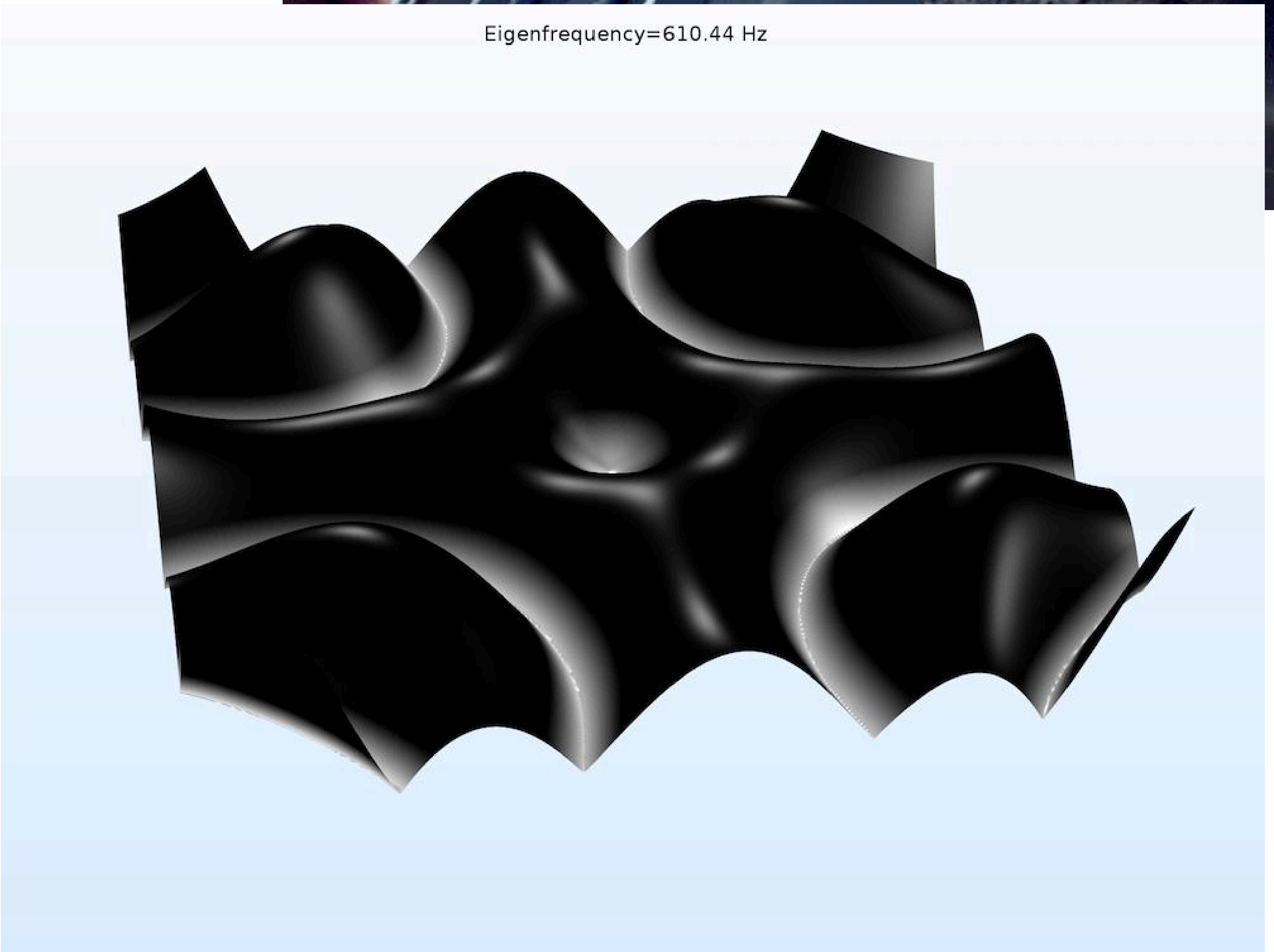
Non-contact, **volumetric structuring** of biomass for organoid growth and replacement?



Background ||

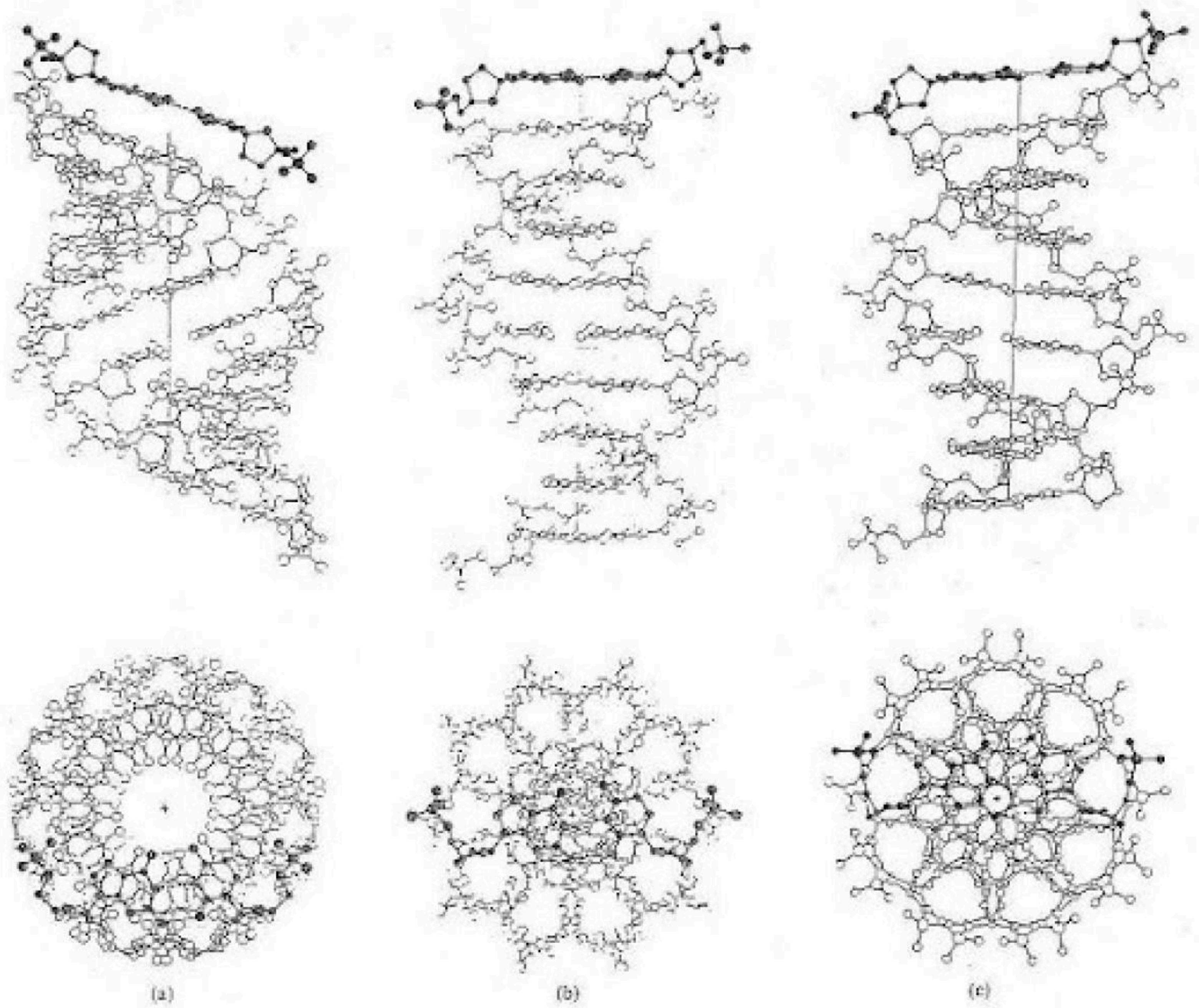
Cymatics is the study of making sound visible by having the acoustic vibrations act upon a medium like sand or water to shape it, allowing sound to express as patterns in matter.



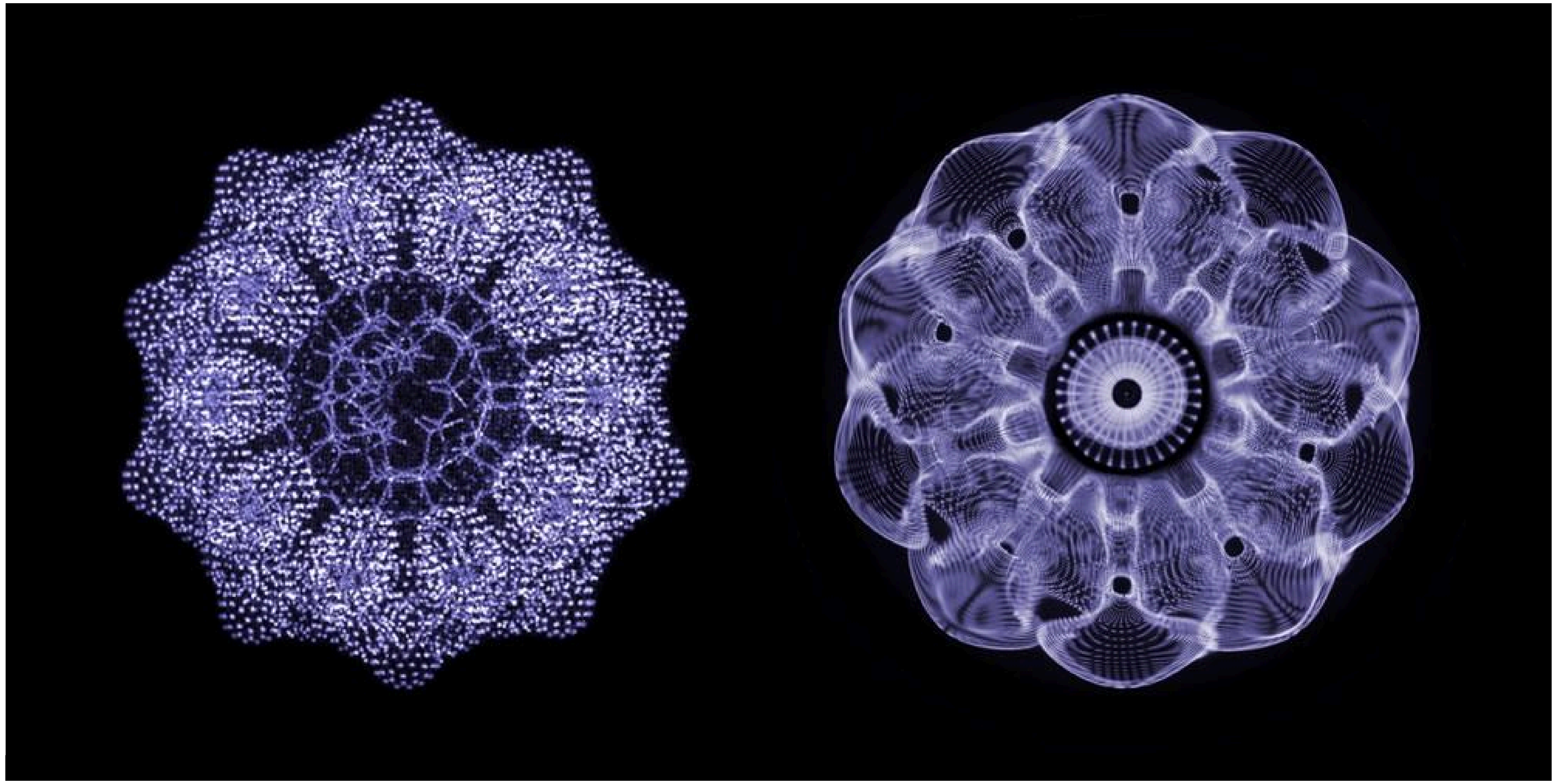



https://www.youtube.com/watch?v=5qmQynxqGjY

When sound interacts with matter, cymatic forces can organise the matter into microscopic and macroscopic structures called "sonic scaffolding".

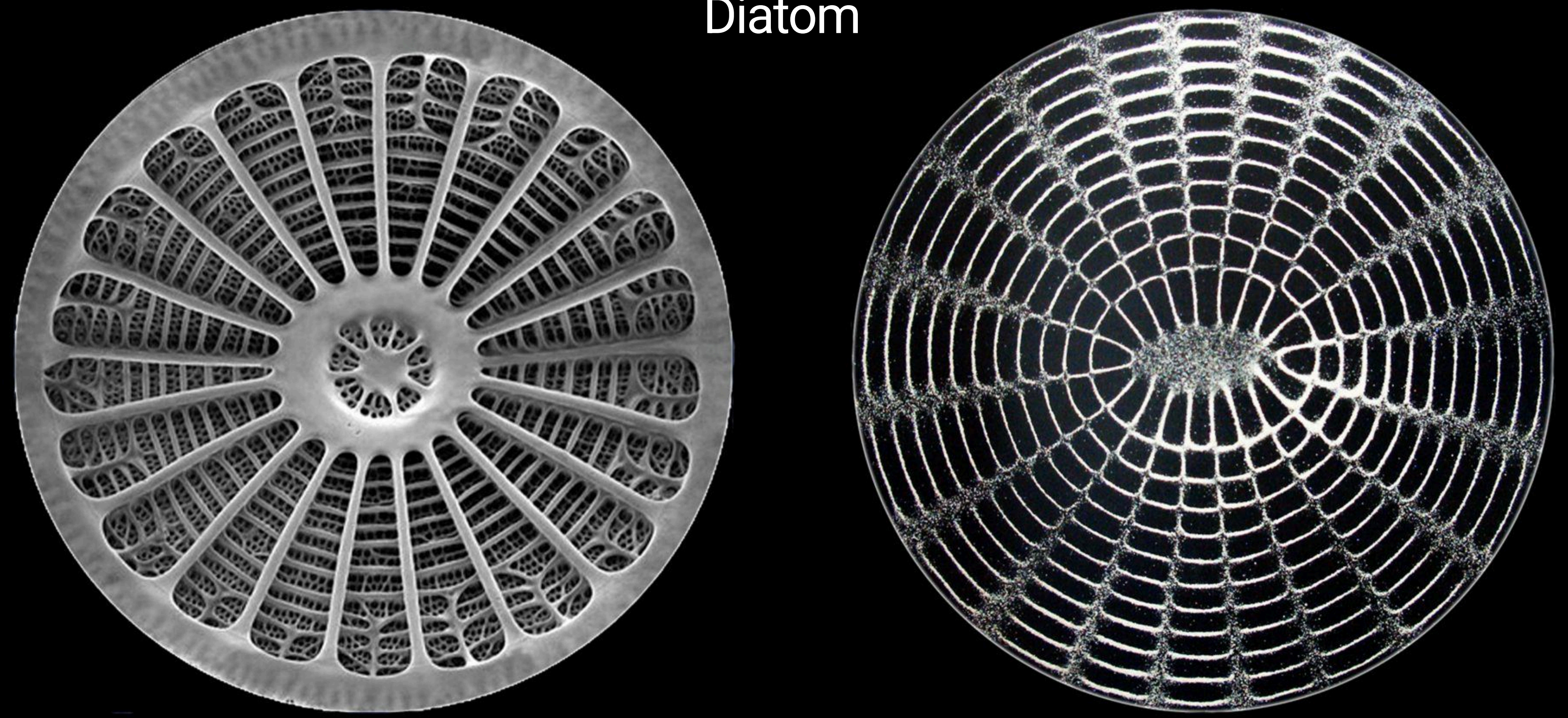


primary structural variations DNA

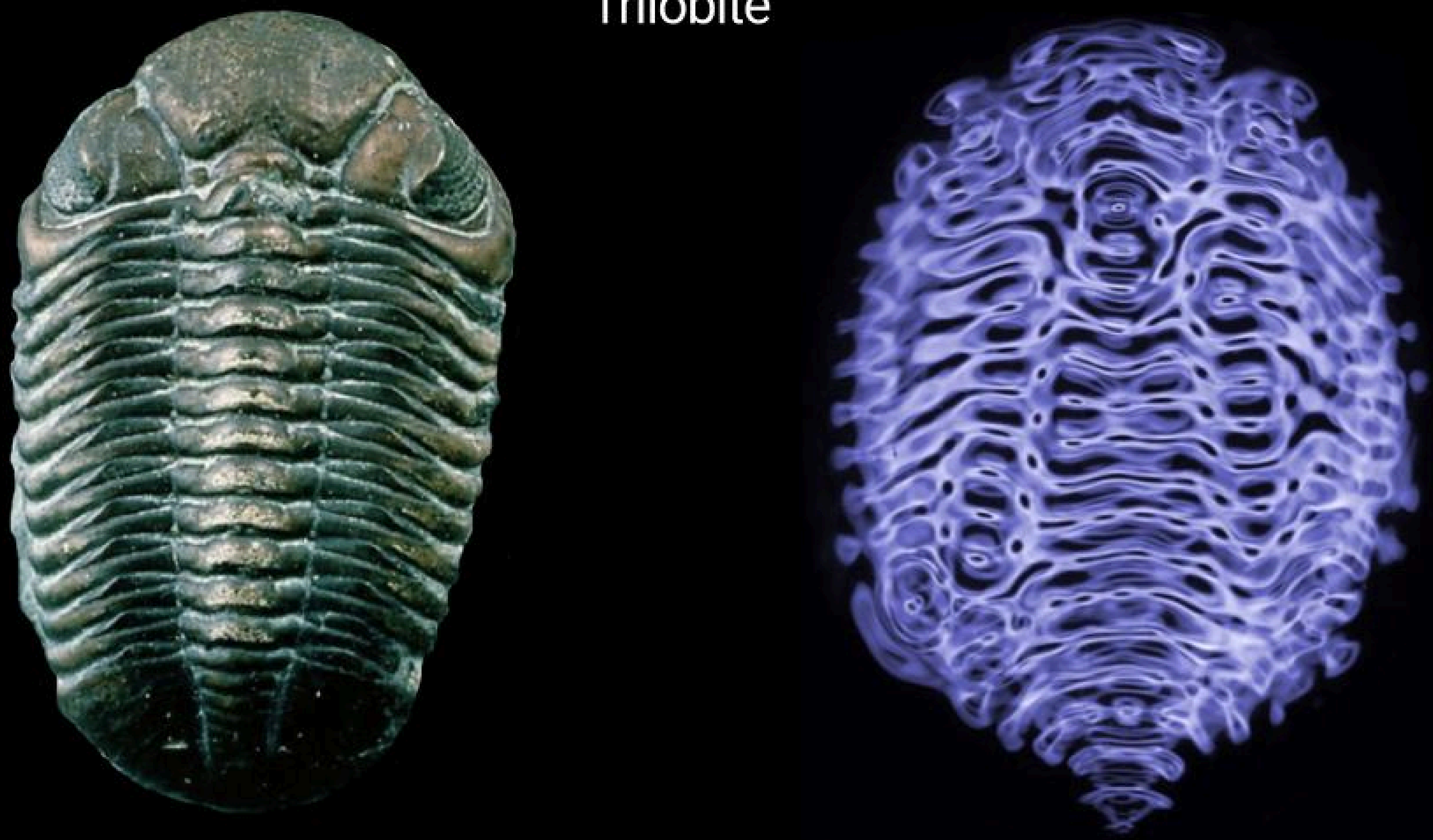


DNA cross-section image vs laboratory-created cymatic

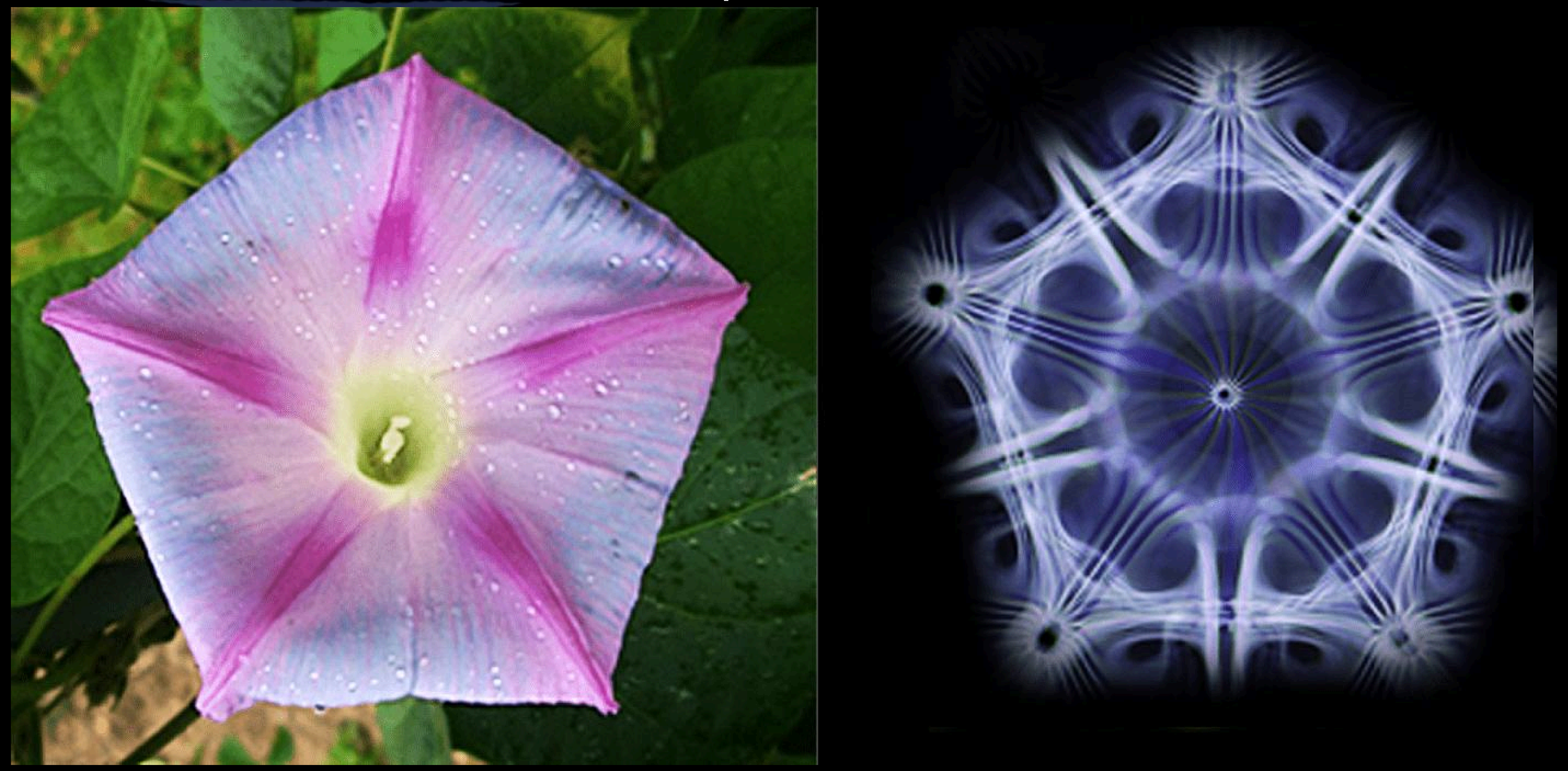
Diatom



Trilobite



Campenula



Starfish

Frequencies that contain certain mathematical ratios appear
to trigger and oragnise biomimetic structures

What can cymatics teach us about how sound shapes biological matter?

Sound Matrix Shaping of Living Matter: From Macrosystems to Cell Microenvironment, Where Mitochondria Act as Energy Portals in Detecting and Processing Sound Vibrations

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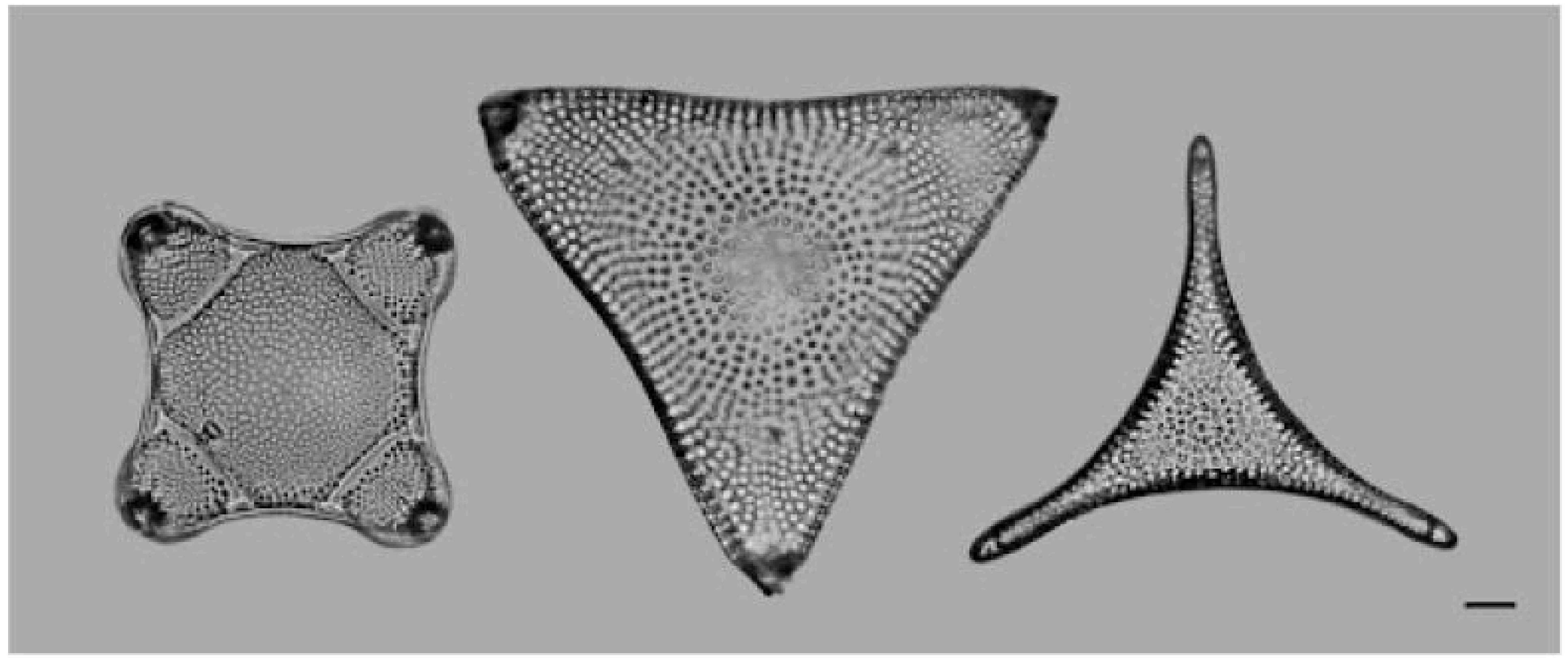
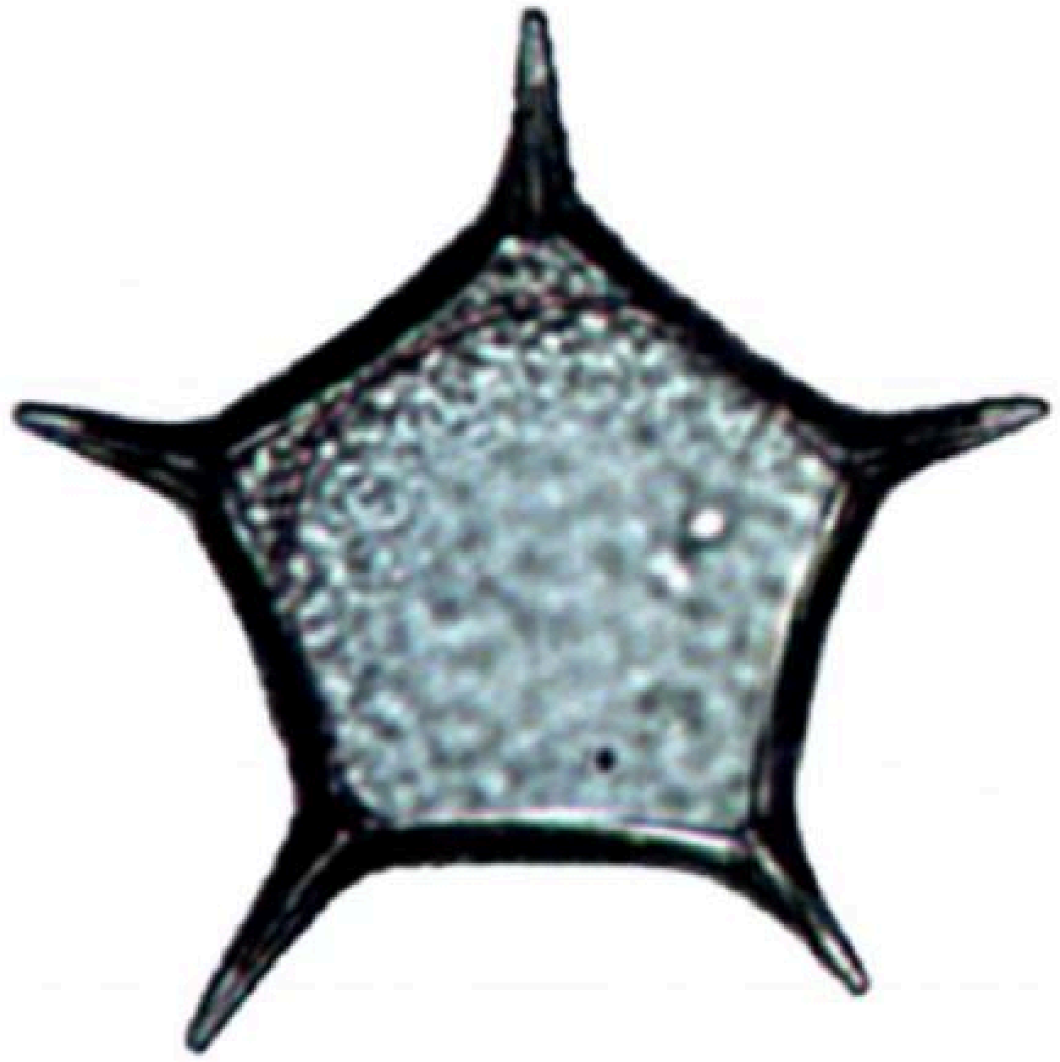
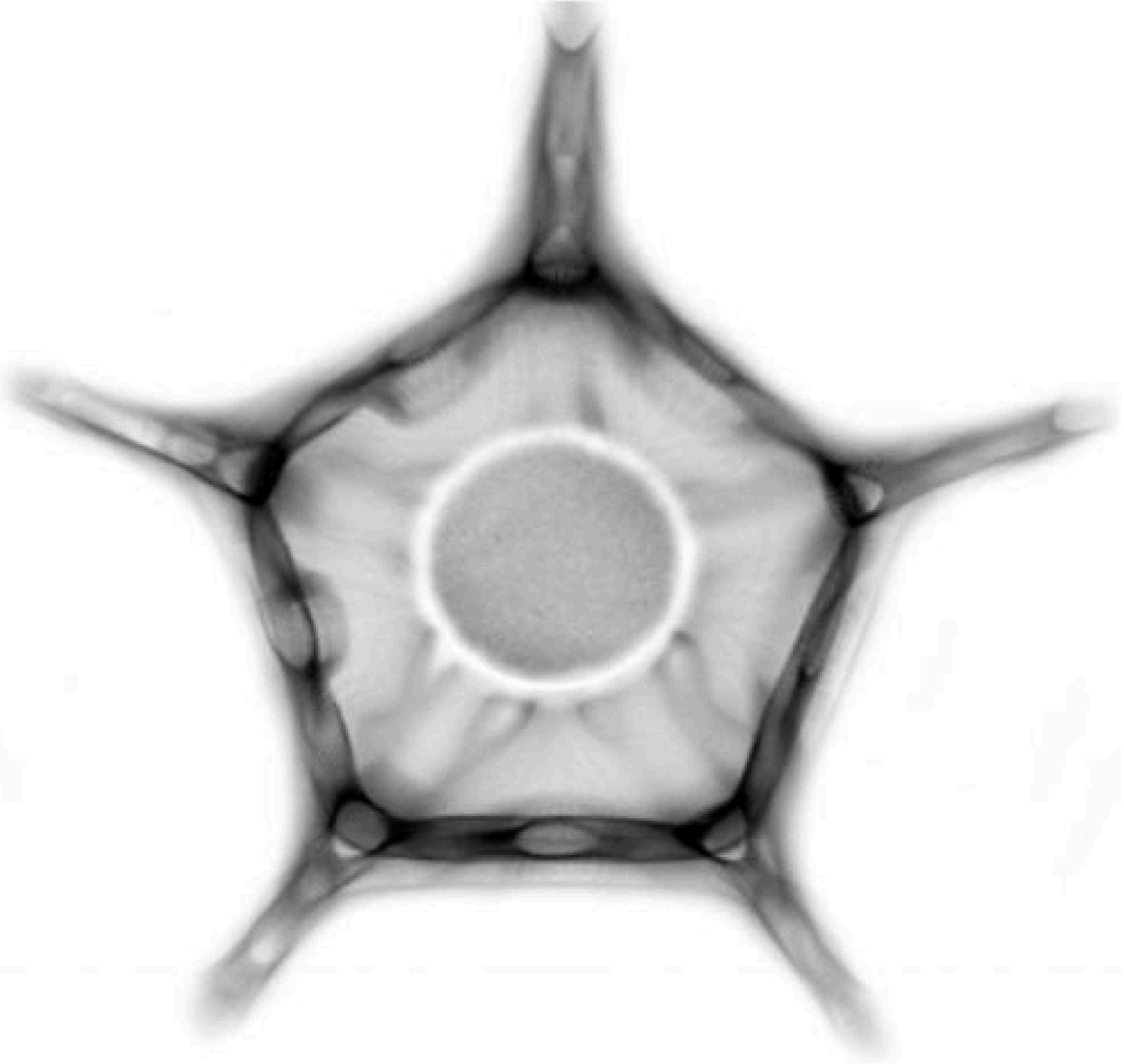
https://www.mdpi.com/1422-0067/25/13/6841

Diatom morphologies



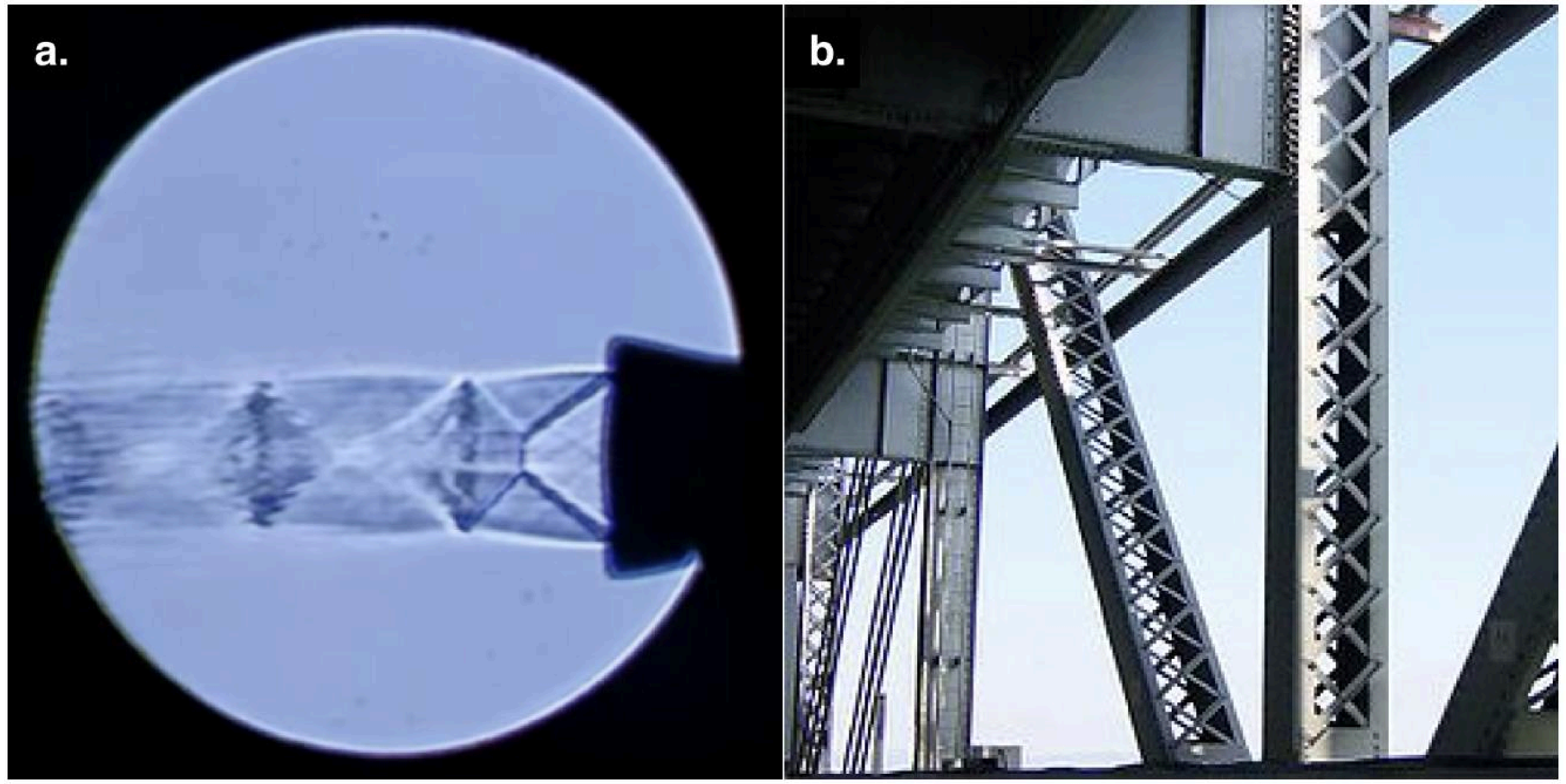




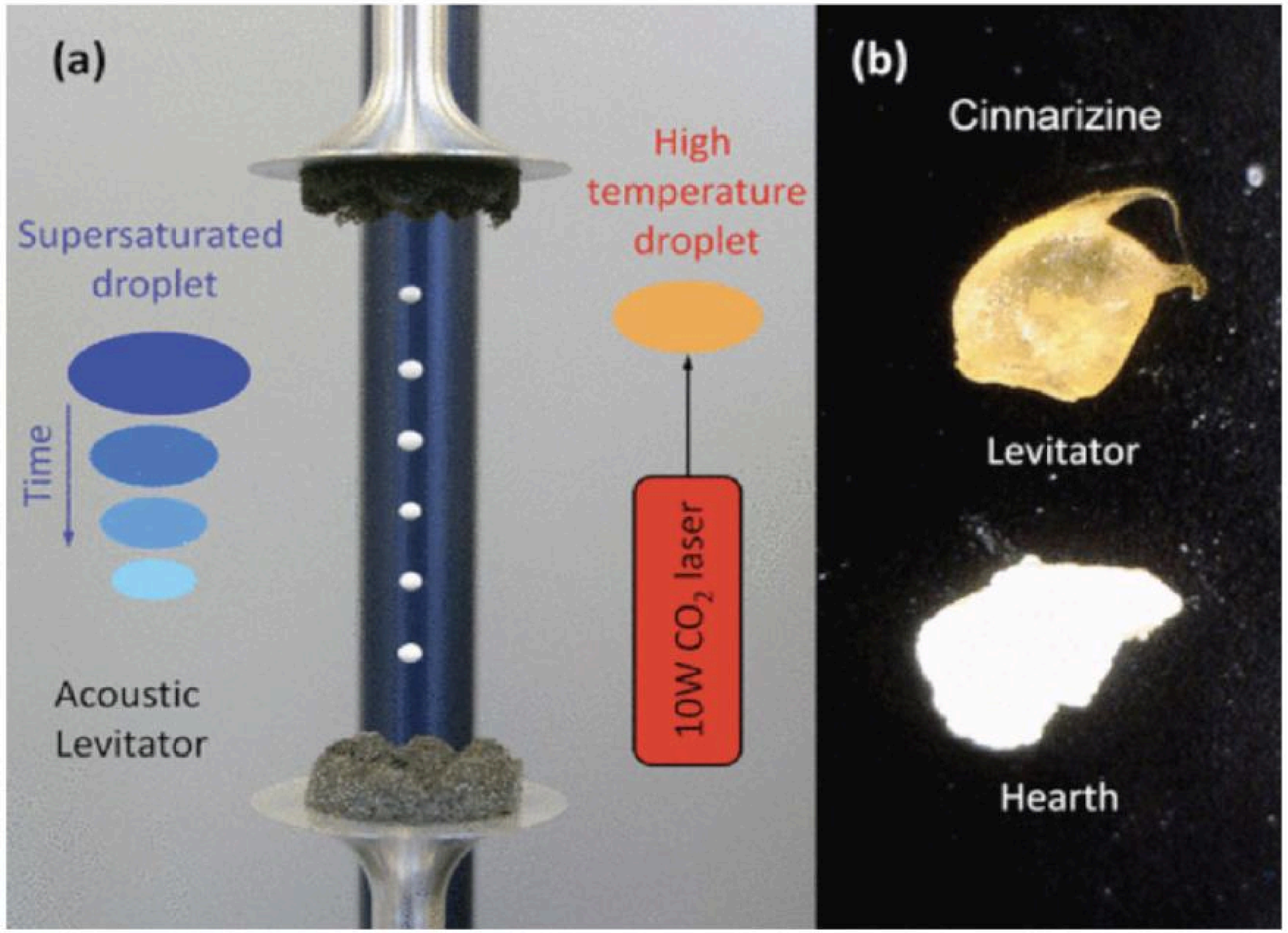
Cretaceous diatom VS cymatic structure

What are the implications for future bio-manufacturing techniques?

Acoustic Morphology of a Large Assembly



A: Schlieren photography reveals cymatic morphologies of pressure fields.
B: an engineered lattice-girder
This parallel illustrates how resonant acoustic geometries may be used ot construct large-scale efficient structures in space



Synthesizing Pharmaceuticals Using Containerless Processing
https://www.anl.gov/partnerships/synthesizing-pharmaceuticals-using-containerless-processing-anlin10011

FAQ / Q&A ||

Why is microgravity useful?
It removes compression effects from gravity, clearer relationship between harmonic and pattern, easier to build delicate structures like artificial organs.

What is cymatics?
Cymatics is the study of visible sound and vibration, from ‘kyma’ - *wave*.

Why biology?
We’re might go a very long way from home using machines, but soft-machines have evolved and with incredible capabilities we should be sure we don’t ignore. Perhaps we can build bits of home into us.